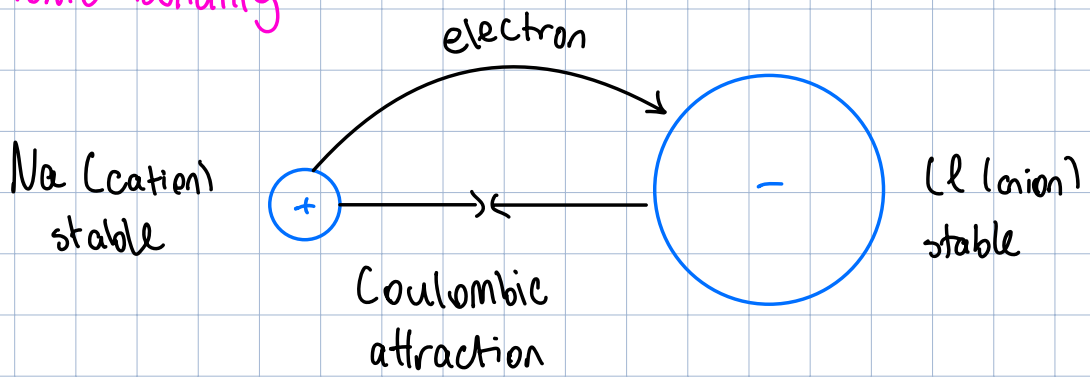
Outline

- types of bonds
  - Ionic
  - Covalent
  - Metallic
  - secondary bonding
- Examples
  - relation between bond E and properties

Ionic Bonding



- between + & − ions
- large diff in electroneg is req
- requires e^- transfer
- typically between metal & non-metal

Typical bonding energies: 600 to 1500 kJ/mole (3 to 8 eV/atom)

typical characteristics of ionically-bonded mats

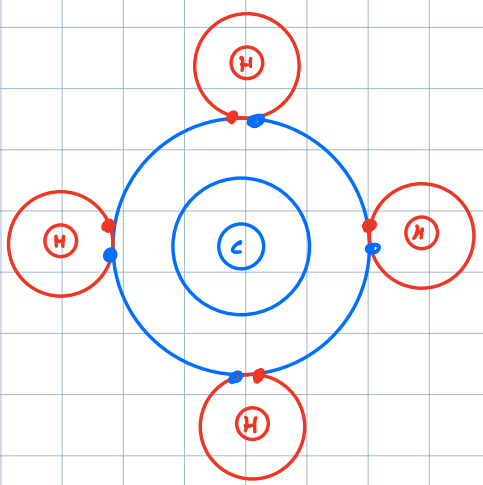
- high melting temp
- hard
- brittle
- insulators (elec & thermal)

Covalent bonding

- req shared e^- ex: CH_4

C: has 4 valence e^-, needs 4 more

H: has 1 ve^-, needs 1 more



- electronegativities are comparable

other ex:

- molecules w/ nonmetals
- molecules w/ metals & non-metal (GaAs)
- Elemental solids (RHS of Periodic table)
- Compound solids (column IV A C, Si, Ge))

summary

- each atom tries to achieve a more stable orbital filling config
- tends to be a highly directional bond
- Gives rise to a fixed orientation of the atoms
- shared e^- may be considered to belong to each atom
- bonds may be strong (diamond, $T_m > 3550°C$) or weak (Bi, $T_m = 270°C$)
- Mats may be electrically conductive (GaAs) or insulating (diamond)

Ionic Character most mats are neither 100% ionic nor 100% covalent

- % ionic character $= \left[1 - \exp\left\{-0.25 (X_A - X_B)^2\right\}\right] \times 100\%$

where X_A and X_B are the elecneg of atom A & B

Metallic Bonding

- Arises from a sea of donated valence e^-

(they're not bounded / free to drift throughout the mat)

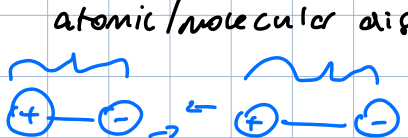
- primary bond for metals & their alloys
- it can be strong or weak
- it gives rise to ↑ elec & thermal conductivity, and ductibility

↓ why?

the $\bar{e}$ are loosely held tg since each atom has many unoccupied valence orbitals; it's relatively easy for the $\bar{e}$ to move about. which allows atoms to slide past eachother

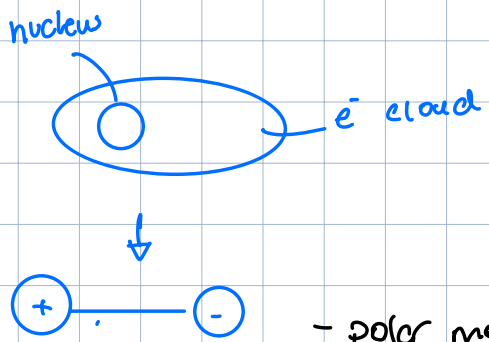
Secondary Bonding

- Van der Waals bonding
- weak compared to others
- exist between almost all atoms / molecules
- Arises from atomic / molecular dipoles



physical bonds, not chemical

↳ (sharing of e^-)



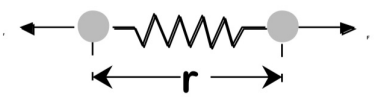
- polar molecule induced dipoles & Hydrogen bonding

↳ the H bond is stronger than the Van der Waals

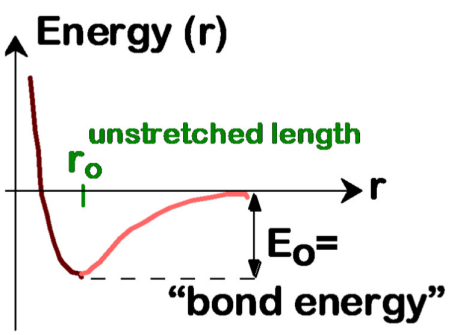


PROPERTIES FROM BONDING: T_M

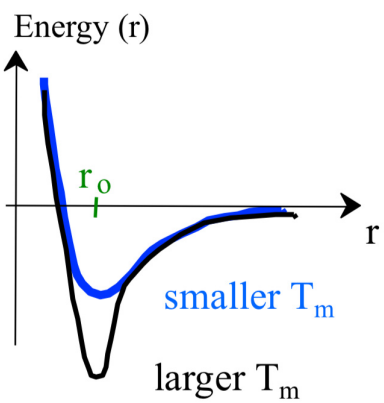
- Bond length, r



- Bond energy, E_o



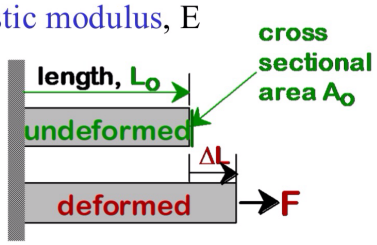
- Melting Temperature, T_m



T_m is larger, if E_o is larger.

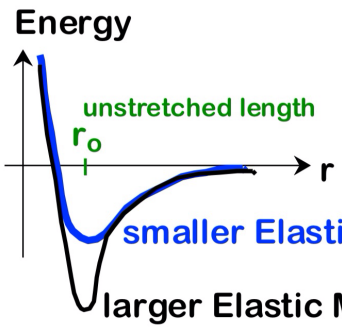
PROPERTIES FROM BONDING: *E*

- Elastic modulus, E



Elastic modulus

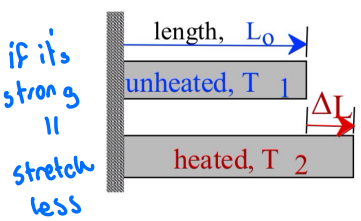
$$\frac{F}{A_o} = E\frac{\Delta L}{L_o}$$



E is larger if E_o is larger.

PROPERTIES FROM BONDING: α

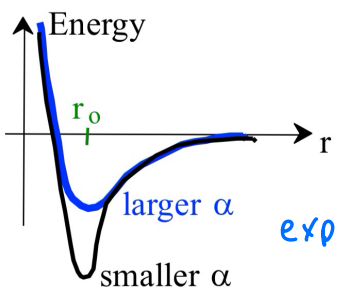
- Coefficient of thermal expansion, α



if it's strong it stretch less

coeff. thermal expansion

$$\frac{\Delta L}{L_o} = \alpha\,(T_2 - T_1)$$



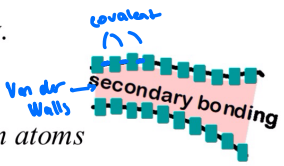
α is larger if E_o is smaller.

expands more when α is larger

Bonding type summary

A comparison of the type of bonding found in different materials:

- For brass, the bonding is metallic since it is a metal alloy.
- For rubber, the bonding is covalent with some V.d.W...
  Rubber is composed primarily of carbon and hydrogen atoms



- For BaS, the bonding is predominantly ionic. (*but with some covalent character*) - *on the basis of the relative positions of Ba and S in the periodic table.*
- For solid xenon, the bonding is Van der Waals since xenon is an inert gas.
- For nylon, the bonding is covalent with perhaps some V.d.W...
  Nylon is composed primarily of carbon and hydrogen
- For AlP the bonding is predominantly covalent (but with some ionic character)
  on the basis of the relative positions of Al and P in the periodic table.

secondary

ex:

Brass (copper + zinc) Cu Zn

2 metals, metallic bonding, alloy

Rubber (C_5H_8)

covalent, N-M, comparable elecneg

Solid Xenon (Xe)

Van der Waals, one element and noble gas

AlP (AlP)

Ionic, M-NM X covalent due to diff in electneg